

Simulation of Random Processes in R

Statistical Computing / Functional Data Analysis

Task 1: Random Walk with Drift

Consider the discrete-time process

$$X_t = X_{t-1} + \mu + \varepsilon_t, \quad \varepsilon_t \sim N(0, \sigma^2), \quad X_0 = 0.$$

Required

1. Simulate $N = 2000$ trajectories for $T = 200$.
2. Plot 50 sample trajectories.
3. Compute empirical:
 - $E[X_t]$
 - $\text{Var}(X_t)$
4. Compare with theoretical values:

$$E[X_t] = \mu t, \quad \text{Var}(X_t) = \sigma^2 t$$

Task 2: AR(1) Process

$$X_t = \phi X_{t-1} + \varepsilon_t, \quad \varepsilon_t \sim N(0, 1)$$

Simulate two cases:

- Stationary: $\phi = 0.7$
- Nonstationary: $\phi = 1$ (unit root)
- Close to nonstationary: $\phi = 0.95$

Required

1. Simulate $T = 2000$
2. Plot the series
3. To do list:
 - plot 50 sample trajectories.
 - compute sample pointwise mean
 - compute sample sample pointwise variance
 - compute sample ACF and PACF for first trajectory
 - estimate ϕ from data for all trajectories

Task 3: Poisson Process

Let $N(t)$ be a Poisson process with rate $\lambda = 2$. Recall that if $N(t)$ is a homogeneous Poisson process with intensity λ , the waiting times between events are independent and exponentially distributed with rate λ . Thus, to simulate $N(t)$ we can sample from the exponential distribution, adding the resulting waiting times together to find the event times.

Required

1. Simulate paths using exponential interarrival times on $[0, 10]$
2. Plot one sample path
3. Compute empirical mean and variance functions
4. Compare with theory:

$$E[N(t)] = \lambda t, \quad \text{Var}(N(t)) = \lambda t$$

R scripts

```
1 #####
2 # Random Processes Lab: 3 Simulation Tasks + Properties
3 #####
4
5 rm(list = ls())
6 set.seed(123)
7
8 # Helper: simple plotting style
9 op <- par(no.readonly = TRUE)
10 on.exit(par(op), add = TRUE)
11
12 #####
13 # TASK 1: Discrete-time random walk with drift
14 #   X_t = X_{t-1} + mu + eps_t, eps_t ~ N(0, sigma^2)
15 #####
16 task1_random_walk <- function(T. = 200, N = 2000, mu = 0.05, sigma = 1) {
17   cat("\n===== \n")
18   cat("TASK 1: Random walk with drift \n")
19   cat("===== \n")
20
21   eps <- replicate(T., rnorm(N, mean = 0, sd = sigma))
22   # cumulative sum across time
23   X <- t(apply(eps, 1, cumsum)) + matrix(mu * rep(1:T., each = N), nrow = N,
24     byrow = FALSE)
25
26   tgrid <- 1:T.
27   emp_mean <- colMeans(X)
28   emp_var <- apply(X, 2, var)
29
30   # Theory (assuming X0 = 0)
31   th_mean <- mu * tgrid
32   th_var <- (sigma^2) * tgrid
33
34   cat(sprintf("Parameters: T=%d, N=%d, mu=%.3f, sigma=%.3f \n", T., N, mu, sigma
35     ))
36   cat("Empirical mean at final time E[X_T]: ", emp_mean[T.], "\n")
37   cat("Theory mean at final time mu*T:      ", th_mean[T.], "\n")
38   cat("Empirical var at final time Var[X_T]:", emp_var[T.], "\n")
39   cat("Theory var at final time sigma^2*T:   ", th_var[T.], "\n")
40 }
```

```

39
40 # Plots
41 par(mfrow = c(2, 2))
42 matplot(tgrid, t(X[1:50, ]), type = "l", lty = 1, xlab = "t", ylab = "X_t",
43         main = "Random walk paths (first 50)")
44
45 plot(tgrid, emp_mean, type = "l", xlab = "t", ylab = "Mean",
46      main = "Empirical vs theory mean")
47 lines(tgrid, th_mean, lty = 2)
48
49 plot(tgrid, emp_var, type = "l", xlab = "t", ylab = "Variance",
50      main = "Empirical vs theory variance")
51 lines(tgrid, th_var, lty = 2)
52
53 invisible(list(X = X, emp_mean = emp_mean, emp_var = emp_var))
54 }
55
56 #####
57 # TASK 2: AR(1) process (stationary vs unit root)
58 # X_t = phi X_{t-1} + eps_t ~ N(0, sigma^2)
59 #####
60 sim_ar1 <- function(T., phi, sigma = 1, x0 = 0) {
61   x <- numeric(T.)
62   x[1] <- x0
63   for (t in 2:T.) x[t] <- phi * x[t - 1] + rnorm(1, 0, sigma)
64   x
65 }
66
67 estimate_phi <- function(x) {
68   fit <- lm(x[-1] ~ x[-length(x)])
69   as.numeric(coef(fit)[2])
70 }
71
72
73 task2_ar1 <- function(T. = 2000, N = 500, sigma = 1, phiA = 0.7, phiB = 1.0,
74                      phiC = 0.95) {
75   cat("\n===== \n")
76   cat("TASK 2: AR(1) stationary vs unit root \n")
77   cat("===== \n")
78
79   xA <- replicate(N, sim_ar1(T., phi = phiA, sigma = sigma))
80   xB <- replicate(N, sim_ar1(T., phi = phiB, sigma = sigma))
81   xC <- replicate(N, sim_ar1(T., phi = phiC, sigma = sigma))
82
83   # Empirical stats
84   stats_one <- function(x, phi_label) {
85     mu_hat <- apply(x, 2, mean)
86     var_hat <- apply(x, 2, var)
87     phi_hat <- apply(x, 2, function(x) estimate_phi(x))
88     cat("\n---", phi_label, "--- \n")
89     cat("Average sample mean: ", mean(mu_hat), "\n")
90     cat("Average sample var: ", mean(var_hat), "\n")
91     cat("Average Phi estimate: ", mean(phi_hat), "\n")
92     invisible(list(mu_hat = mu_hat, var_hat = var_hat, phi_hat = phi_hat))
93   }
94
95   outA <- stats_one(xA, sprintf("Case A (stationary): phi=%.2f", phiA))
96   outB <- stats_one(xB, sprintf("Case B (unit root): phi=%.2f", phiB))
97   outC <- stats_one(xC, sprintf("Case C (near unit root): phi=%.2f", phiC))
98
99   # Plots
100  par(mfrow = c(3, 3))

```

```

101  matplot(xA[,1:20], type = "l", main = sprintf("AR(1) series (phi=%.2f)", phiA)
    , xlab = "t", ylab = "X_t")
102  matplot(xB[,1:20], type = "l", main = sprintf("AR(1) series (phi=%.2f)", phiB)
    , xlab = "t", ylab = "X_t")
103  matplot(xC[,1:20], type = "l", main = sprintf("AR(1) series (phi=%.2f)", phiC)
    , xlab = "t", ylab = "X_t")
104
105  acf(xA[,1], main = sprintf("ACF (phi=%.2f)", phiA))
106  acf(xB[,1], main = sprintf("ACF (phi=%.2f)", phiB))
107  acf(xC[,1], main = sprintf("ACF (phi=%.2f)", phiC))
108
109  pacf(xA[,1], main = sprintf("PACF (phi=%.2f)", phiA))
110  pacf(xB[,1], main = sprintf("PACF (phi=%.2f)", phiB))
111  pacf(xC[,1], main = sprintf("PACF (phi=%.2f)", phiC))
112
113  par(mfrow = c(3, 3))
114  boxplot(outA$mu_hat, main = sprintf("Sample mean (phi=%.2f)", phiA), xlab = "t"
    , ylab = "X_t")
115  boxplot(outB$mu_hat, main = sprintf("Sample mean (phi=%.2f)", phiB), xlab = "t"
    , ylab = "X_t")
116  boxplot(outC$mu_hat, main = sprintf("Sample mean (phi=%.2f)", phiC), xlab = "t"
    , ylab = "X_t")
117
118  boxplot(outA$var_hat, main = sprintf("Sample variance (phi=%.2f)", phiA), xlab
    = "t", ylab = "X_t")
119  boxplot(outB$var_hat, main = sprintf("Sample variance (phi=%.2f)", phiB), xlab
    = "t", ylab = "X_t")
120  boxplot(outC$var_hat, main = sprintf("Sample variance (phi=%.2f)", phiC), xlab
    = "t", ylab = "X_t")
121
122  boxplot(outA$phi_hat, main = sprintf("Coefficient estimates (phi=%.2f)", phiA)
    , xlab = "t", ylab = "X_t")
123  boxplot(outB$phi_hat, main = sprintf("Coefficient estimates (phi=%.2f)", phiB)
    , xlab = "t", ylab = "X_t")
124  boxplot(outC$phi_hat, main = sprintf("Coefficient estimates (phi=%.2f)", phiC)
    , xlab = "t", ylab = "X_t")
125
126
127  invisible(list(xA = xA, xB = xB, outA = outA, outB = outB))
128 }
129
130 #####
131 # TASK 3: Poisson process
132 #   Inter-arrival times ~ Exp(lambda)
133 #   N(t) counts arrivals up to time t
134 #####
135 sim_pois_proc_arrivals <- function(Tmax, lambda) {
136   t <- 0
137   times <- numeric(0)
138   repeat {
139     t <- t + rexp(1, rate = lambda)
140     if (t > Tmax) break
141     times <- c(times, t)
142   }
143   times
144 }
145
146 counts_on_grid <- function(arrivals, grid) {
147   sapply(grid, function(g) sum(arrivals <= g))
148 }
149
150 task3_poisson_process <- function(Tmax = 10, lambda = 2, dt = 0.1, M = 2000) {
151   cat("\n===== \n")

```

```

152 cat("TASK 3: Poisson process\n")
153 cat("=====\n")
154
155 grid <- seq(0, Tmax, by = dt)
156 Nt_mat <- matrix(0, nrow = M, ncol = length(grid))
157
158 for (m in 1:M) {
159   arr <- sim_pois_proc_arrivals(Tmax, lambda)
160   Nt_mat[m, ] <- counts_on_grid(arr, grid)
161 }
162
163 emp_mean <- colMeans(Nt_mat)
164 emp_var <- apply(Nt_mat, 2, var)
165
166 th_mean <- lambda * grid
167 th_var <- lambda * grid
168
169 cat(sprintf("Parameters: Tmax=%.2f, dt=%.2f, M=%d, lambda=%.3f\n", Tmax, dt, M
170   , lambda))
171 cat("At final time:\n")
172 cat(" Empirical E[N(T)]: ", emp_mean[length(grid)], "\n")
173 cat(" Theory E[N(T)]: ", th_mean[length(grid)], "\n")
174 cat(" Empirical Var[N(T)]: ", emp_var[length(grid)], "\n")
175 cat(" Theory Var[N(T)]: ", th_var[length(grid)], "\n")
176
177 # Plots
178 par(mfrow = c(1, 2))
179 plot(grid, emp_mean, type = "l", xlab = "t", ylab = "E[N(t)]",
180   main = "Mean function: empirical vs theory")
181 lines(grid, th_mean, lty = 2)
182
183 plot(grid, emp_var, type = "l", xlab = "t", ylab = "Var[N(t)]",
184   main = "Variance function: empirical vs theory")
185 lines(grid, th_var, lty = 2)
186
187 invisible(list(Nt_mat = Nt_mat, grid = grid))
188 }
189
190 #####
191 # RUN ALL TASKS
192 #####
193 res1 <- task1_random_walk(T. = 200, N = 2000, mu = 0.05, sigma = 1)
194 res2 <- task2_ar1(T. = 2000, N = 500, sigma = 1, phiA = 0.7, phiB = 1.0, phiC =
195   0.95)
196 res3 <- task3_poisson_process(Tmax = 10, lambda = 2, dt = 0.1, M = 2000)

```